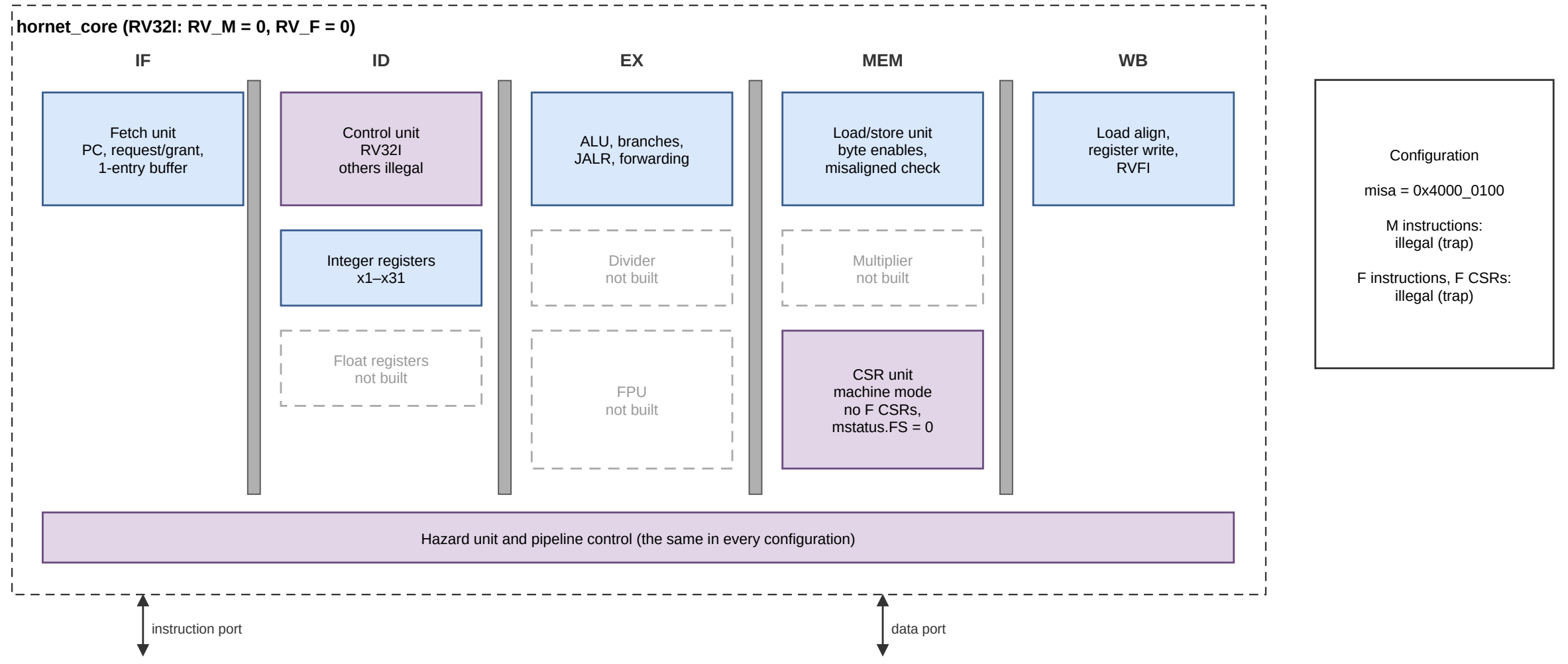
hornet_core (RV32I: RV_M = 0, RV_F = 0)
IF
ID
EX
MEM
WB
Fetch unit
PC, request/grant,
1-entry buffer
Control unit
RV32I
others illegal
Integer registers
x1–x31
Float registers
not built
ALU, branches,
JALR, forwarding
Divider
not built
FPU
not built
Load/store unit
byte enables,
misaligned check
Multiplier
not built
CSR unit
machine mode
no F CSRs,
mstatus.FS = 0
Load align,
register write,
RVFI
Hazard unit and pipeline control (the same in every configuration)
Configuration
misa = 0x4000_0100
M instructions:
illegal (trap)
F instructions, F CSRs:
illegal (trap)
instruction port
data port